

Problem-Solution Fit

Date	3 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

#	Category	Description
1	Customer Segment (CS)	Commercial banks, NBFCs, credit card issuers, and credit analysts who process thousands of applications monthly
2	Problem (PR)	Manual credit card approval is slow, inconsistent, prone to human bias, and fails to leverage the patterns hidden in historical credit data
3	Target Result (TR)	An ML-powered prediction tool that instantly classifies an applicant as Approved or Rejected based on 15 objective features
4	Expected Measurement (EM)	Prediction delivered within 2 seconds with a confidence score %; Random Forest achieves 81.91% test accuracy and 0.89 ROC-AUC
5	Adopted Solution (AS)	Flask web application serving a trained Random Forest classifier, with preprocessing pipeline (LabelEncoder + StandardScaler) saved as PKL artifacts
6	Collected	Training on Kaggle's real-world dataset (438K+

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	Lessons (CL)	applicants, 1M+ monthly records) validated that employment status, income, and credit history are the strongest predictors
7	Business Enabler (BE)	Reduces manual review time by ~90%; enables credit analysts to handle higher application volumes without proportional headcount increase
8	Channel (CH)	Deployed at https://credit-card-approval-prediction-ra22.onrender.com via Render; auto-deploys on every GitHub push to master
9	Revenue / Cost (RC)	Zero infrastructure cost using Render free tier + GitHub + open-source Python stack
10	Scalability Level (SL)	Stateless Flask app scales horizontally; model can be retrained with updated data and redeployed without UI changes